Final Project
CSC 453
Intro to OS

CAL POLY
SAN LUIS OBISPO

The Final Assignment

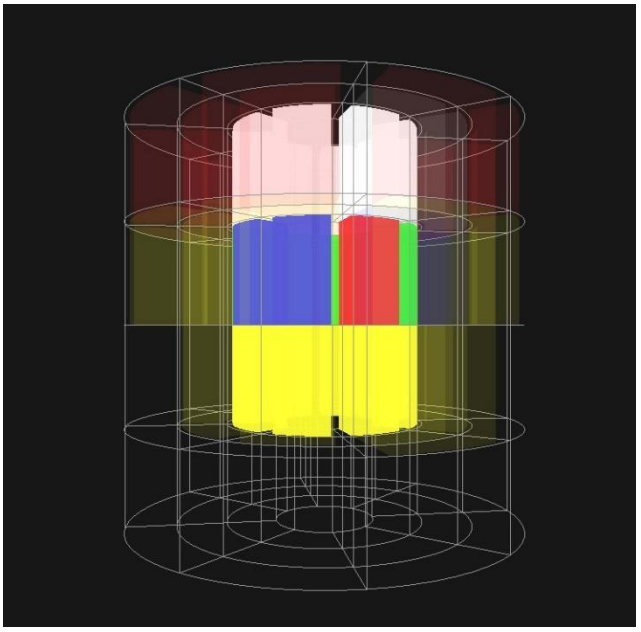


Figure 1: Your visualizer in action

Create a File System

Create a file system, this task is independent of your OS. The filesystem will cater to a 4-platter HDD.

Lets clarify terminology first

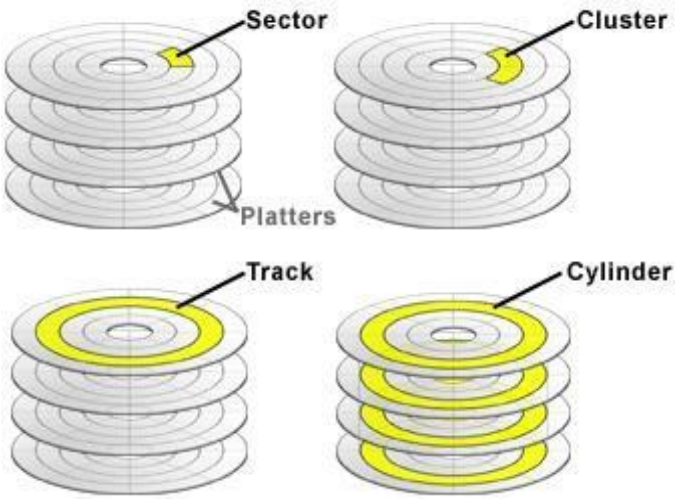


Figure 2: A multi-platter HDD

This will be our hard drive. It has 4 platters. The 4 magnetic read/write heads must go simultaneously to the desired track. This is how HDD are built. The heads cannot move independently. That means file blocks for one file are preferred along the vertical axis on the same sector-cylinder.

Task – Whats needs to be done

Read commands from the input file `input.txt`.

```
CREATE 2 4
CREATE 3 3
CREATE 4 4
DELETE 3
CREATE 5 50
```

read/write into a harddrive-file which represents our harddrive. Those files (i.e. HDD) should be named `step_0.bin`, `step_1.bin`, etc.

We will test your program with 10 commands. Each command should lead to a different output file.

The attached visualizer can read these harddrive files and show you if you are successful.

HDD and File Blocks:

Our harddrive will have

- Four Platters
- Three tracks per platter
- Eight sectors per track, see fig 1 and 2.

Each sector holds one block. One Block is 256 bytes.

The enumeration works from top platter to bottom, starting concentric from the 8 sectors of the inner track to the middle track to the outer track and then to the next platter below etc. 8 sectors, 3 tracks, 4 platters ... 96 blocks in sum.

What the file system must have:

Store/write/read/remove files on the HDD.

A superblock should be block 0 and it should keep track of the free blocks. The first byte of this block must be a `1` as indication that it is the superblock.

For each file, the first block can be seen as the inode and must contain the indices of the other blocks (indexed allocation). Files are labelled by numbers and not names. The first byte of a block that belongs to a file must have that exact number, i.e. the first byte of **each** block must contain the file number (2-9, we only have a max of 8 files). Our files simply have numbers from 2-9 instead of names.

What your file system must do:

Write file blocks vertically! For example, a file with 4 blocks should be on the same sector in the same cylinder on all 4 platters. If it needs to be split, just continue on the next sector and go vertical from there.

Input and Output Files

Attached is a sample input file with 5 inputs. You will be tested against 10 inputs once submitted. Each successful output will be 10% of your grade.

Inputs can be:

CREATE 2 4 //means create a new file, named 2, usage 4 blocks

DELETE 3 //means delete file number 3

After each command, create a new harddrive file.

These files have to be named `step_0.bin` for the first command and `step_1.bin` for the second, etc.

These binary output file must contain all bytes of your harddrive file: 96 x 256 bytes! All blocks! Regardless of empty or not:

Block	**First byte**
Superblock	1
Fileblock	2-9
Empty	0

The Visualizer Program

Your visualizer program lets you see what you did in the binary harddrive files, see fig. 1. Your step-files must be in the same folder. Once started, you will see the HDD in front of you with the information from the first step. If a step file is missing, the ones existing are still loaded.

Key bindings:

WASD … move camera

Q,E … move camera up down

0 – 9 ... move through your step files

Z,X,C,SPACE … takes a closer look on each cylinder and makes the others more transparent. It can get complicated! SPACE is resetting this setting.

ESC … quits the program.

Submit

Submit everything zipped with your name and last name as file name to Canvas!

Good Luck.